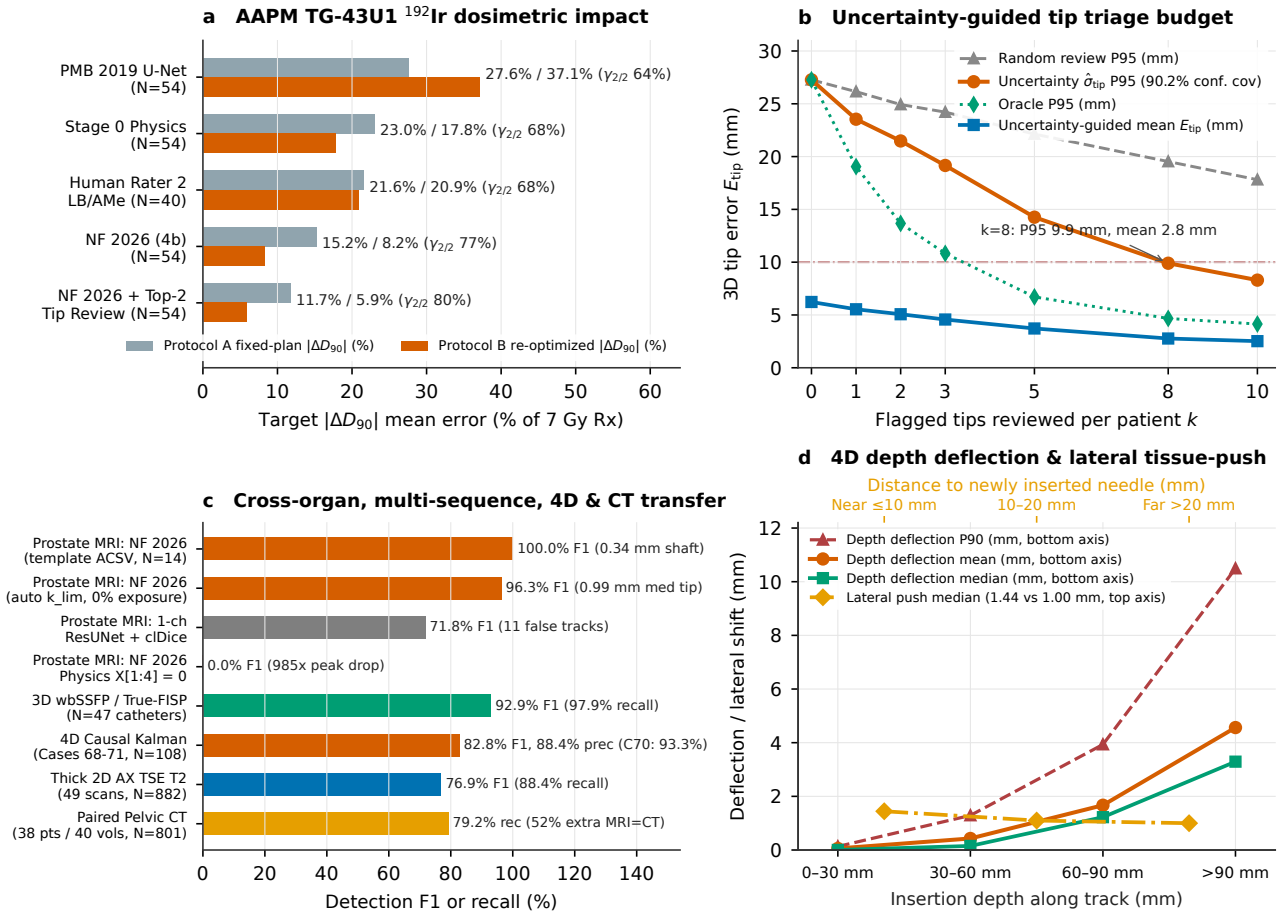
a AAPM TG-43U1 192Ir dosimetric impact
PMB 2019 U-Net (N=54)
27.6% / 37.1% (γ2/2 64%)
Stage 0 Physics (N=54)
23.0% / 17.8% (γ2/2 68%)
Human Rater 2 LB/AMe (N=40)
21.6% / 20.9% (γ2/2 68%)
NF 2026 (4b) (N=54)
15.2% / 8.2% (γ2/2 77%)
NF 2026 + Top-2 Tip Review (N=54)
11.7% / 5.9% (γ2/2 80%)
Protocol A fixed-plan |ΔD90| (%)
Protocol B re-optimized |ΔD90| (%)
0 10 20 30 40 50 60
Target |ΔD90| mean error (% of 7 Gy Rx)
b Uncertainty-guided tip triage budget
Random review P95 (mm)
Uncertainty σ̂tip P95 (90.2% conf. cov)
Oracle P95 (mm)
Uncertainty-guided mean Etip (mm)
k=8: P95 9.9 mm, mean 2.8 mm
3D tip error Etip (mm)
0 5 10 15 20 25 30
0 1 2 3 5 8 10
Flagged tips reviewed per patient k
c Cross-organ, multi-sequence, 4D & CT transfer
Prostate MRI: NF 2026 (template ACSV, N=14)
100.0% F1 (0.34 mm shaft)
Prostate MRI: NF 2026 (auto k_lim, 0% exposure)
96.3% F1 (0.99 mm med tip)
Prostate MRI: 1-ch ResUNet + clDice
71.8% F1 (11 false tracks)
Prostate MRI: NF 2026 Physics X[1:4] = 0
0.0% F1 (985x peak drop)
3D wbSSFP / True-FISP (N=47 catheters)
92.9% F1 (97.9% recall)
4D Causal Kalman (Cases 68-71, N=108)
82.8% F1, 88.4% prec (C70: 93.3%)
Thick 2D AX TSE T2 (49 scans, N=882)
76.9% F1 (88.4% recall)
Paired Pelvic CT (38 pts / 40 vols, N=801)
79.2% rec (52% extra MRI=CT)
0 20 40 60 80 100 120 140
Detection F1 or recall (%)
d 4D depth deflection & lateral tissue-push
Distance to newly inserted needle (mm)
Near ≤10 mm
10–20 mm
Far >20 mm
Depth deflection P90 (mm, bottom axis)
Depth deflection mean (mm, bottom axis)
Depth deflection median (mm, bottom axis)
Lateral push median (1.44 vs 1.00 mm, top axis)
Deflection / lateral shift (mm)
0 2 4 6 8 10 12
0–30 mm
30–60 mm
60–90 mm
>90 mm
Insertion depth along track (mm)